

Enterprise Cryptographic Discovery & Analysis Tool

Scan Report -- ID a1c27bb3 -- generated 2026-09-26T17:46:38.294287+00:00

Metric	Value
Total artefacts scanned	14
Crypto-Agility Score	37.5 / 100 (Grade D, Level 1 -- Initial)
CRQC planning horizon (Z, years)	12
Critical-risk artefacts	5
High-risk artefacts	0
Harvest-now-decrypt-later exposure	480.0 GB

Assumption disclosure: the CRQC planning horizon (Z) above is a configurable planning assumption, not a verified prediction -- see README for its basis. Regulatory framework flags in this report are automated triage suggestions, not legal determinations.

1. Cryptographic Asset Inventory

Name	Algorithm	Type	Occurrences	Risk Tier	Confidence
api-gateway-tls-cert	RSA-2048	certificate	1	Critical	high
auth-service-jwt-signing-key	ECDSA-P256	key	1	Critical	high
payments-db-encryption	AES-256-GCM	algorithm	1	Safe	high
legacy-file-transfer-cipher	3DES	algorithm	1	Low	high
openssl-libcrypto	RSA-4096	library	1	Critical	high
session-key-exchange	ECDH-P384	protocol	1	Critical	high
internal-log-signing-key	DSA-2048	key	1	Medium	high
next-gen-vpn-handshake	ML-KEM-768	protocol	1	Safe	high
config-secrets-cipher	AES-128-CBC	algorithm	3	Low	high
partner-api-mutual-tls	ECDSA-P256	certificate	1	Critical	high
code-signing-key	RSA-3072	key	1	Medium	high
document-signature-service	ML-DSA-65	algorithm	1	Safe	high
internal-admin-tls-cert	RSA-2048	certificate	1	Medium	high
possible-des-reference-in-comment	DES	algorithm	1	Low	low

2. Risk Assessment & Recommendations

Name	Current	Recommended Replacement	Priority
api-gateway-tls-cert	RSA-2048	ML-KEM-768 (key exchange) + ML-DSA-65 (signatures), or hybrid X25519+ML-KEM-768 during transition	Critical
auth-service-jwt-signing-key	ECDSA-P256	ML-DSA-65, or hybrid ECDSA+ML-DSA during transition	Critical
payments-db-encryption	AES-256-GCM	AES-256-GCM	Safe
legacy-file-transfer-cipher	3DES	AES-256-GCM	Low
openssl-libcrypto	RSA-4096	ML-KEM-768 (key exchange) + ML-DSA-65 (signatures), or hybrid X25519+ML-KEM-768 during transition	Critical
session-key-exchange	ECDH-P384	ML-KEM-768, or hybrid X25519+ML-KEM-768	Critical
internal-log-signing-key	DSA-2048	ML-DSA-65 (Dilithium)	Medium
next-gen-vpn-handshake	ML-KEM-768	ML-KEM-768	Safe
config-secrets-cipher	AES-128-CBC	AES-256 (same algorithm, upgraded key length)	Low
partner-api-mutual-tls	ECDSA-P256	ML-DSA-65, or hybrid ECDSA+ML-DSA during transition	Critical
code-signing-key	RSA-3072	ML-KEM-768 (key exchange) + ML-DSA-65 (signatures), or hybrid X25519+ML-KEM-768 during transition	Medium
document-signature-service	ML-DSA-65	ML-DSA-65	Safe
internal-admin-tls-cert	RSA-2048	ML-KEM-768 (key exchange) + ML-DSA-65 (signatures), or hybrid X25519+ML-KEM-768 during transition	Medium
possible-des-reference-in-comment	DES	AES-256-GCM	Low

3. Phased Migration Roadmap

Phase 1 (0-6 months) -- 5 artefact(s)

- session-key-exchange (Crown Jewel): ECDH-P384 → ML-KEM-768, or hybrid X25519+ML-KEM-768
- api-gateway-tls-cert (Crown Jewel): RSA-2048 → ML-KEM-768 (key exchange) + ML-DSA-65 (signatures), or hybrid X25519+ML-KEM-768 during transition
- auth-service-jwt-signing-key (Crown Jewel): ECDSA-P256 → ML-DSA-65, or hybrid ECDSA+ML-DSA during transition
- openssl-libcrypto (Crown Jewel): RSA-4096 → ML-KEM-768 (key exchange) + ML-DSA-65 (signatures), or hybrid X25519+ML-KEM-768 during transition
- partner-api-mutual-tls (Crown Jewel): ECDSA-P256 → ML-DSA-65, or hybrid ECDSA+ML-DSA during transition

Phase 2 (6-18 months) -- 0 artefact(s)

Phase 3 (18-36 months) -- 3 artefact(s)

- internal-log-signing-key (Standard): DSA-2048 → ML-DSA-65 (Dilithium)
- code-signing-key (Standard): RSA-3072 → ML-KEM-768 (key exchange) + ML-DSA-65 (signatures), or hybrid X25519+ML-KEM-768 during transition
- internal-admin-tls-cert (Standard): RSA-2048 → ML-KEM-768 (key exchange) + ML-DSA-65 (signatures), or hybrid X25519+ML-KEM-768 during transition

Phase 4 (36+ months / monitor) -- 3 artefact(s)

- legacy-file-transfer-cipher (Standard): 3DES → AES-256-GCM
- config-secrets-cipher (Standard): AES-128-CBC → AES-256 (same algorithm, upgraded key length)
- possible-des-reference-in-comment (Standard): DES → AES-256-GCM

4. Regulatory / Compliance Triage Flags

Automated triage suggestions only -- confirm with a compliance/legal reviewer.

- **art-001:** DPDP Act 2023, CERT-In Directions 2022, NCIIPC Guidelines
- **art-002:** DPDP Act 2023, CERT-In Directions 2022, NCIIPC Guidelines
- **art-003:** RBI Cybersecurity Framework, SEBI Cybersecurity Framework
- **art-005:** RBI Cybersecurity Framework, SEBI Cybersecurity Framework, CERT-In Directions 2022, NCIIPC Guidelines
- **art-006:** DPDP Act 2023, CERT-In Directions 2022, NCIIPC Guidelines
- **art-008:** RBI Cybersecurity Framework, SEBI Cybersecurity Framework, CERT-In Directions 2022, NCIIPC Guidelines
- **art-010:** RBI Cybersecurity Framework, SEBI Cybersecurity Framework, CERT-In Directions 2022, NCIIPC Guidelines
- **art-012:** DPDP Act 2023, CERT-In Directions 2022, NCIIPC Guidelines